

Phase 8 – Project Demonstration

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

8.1 Introduction

The Project Demonstration phase presents the completed ServiceNow application by demonstrating each milestone of the implementation. The demonstration verifies that Script-Controlled Access Control Lists (ACLs) successfully restrict record access based on user roles and the Branch field value while administrators retain full access.

Milestone 1 – Creation of Users and Roles

User Creation

1. Log in to the ServiceNow instance with Admin access.
2. Navigate to User Administration → Users.
3. Click New.
4. Create the EEE User with the following details:
 - a. User ID: EEE User
 - b. First Name: EEE
 - c. Last Name: User
 - d. Email: eeeuser@gmail.com
5. Save the record.

Screenshot: EEE User Record.

The screenshot shows the ServiceNow User Administration interface for a user named 'EEE User'. The page is divided into two main sections: user details on the left and system settings on the right. The user details section includes fields for User ID (EEE User), First name (EEE), Last name (User), Title, and Department. There are also checkboxes for 'Password needs reset', 'Locked out', and 'Active' (which is checked). A dropdown menu for 'Identity type' is set to 'Human'. The system settings section includes fields for Email (eeeuser@gmail.com), Language (None), Calendar integration (Outlook), Time zone (System (America/Los Angeles)), Date format (System (2026-08-19)), Business phone, and Mobile phone. There is also a 'Photo' field with a 'Click to add...' link. At the bottom of the form, there are buttons for 'Update', 'Set Password', and 'Delete'. Below the form, there are 'Related Links' for 'View linked accounts', 'View Subscriptions', and 'Reset a password'. The bottom of the screenshot shows a Windows taskbar with various application icons and a system clock indicating 19:15 on 27-09-2026.

servicenow All Favorites History Workspaces User - EEE User Search

User ID: EEE User
First name: EEE
Last name: User
Title:
Department:
Password needs reset: ☐
Locked out: ☐
Active: ☒
Identity type: Human
Internal Integration User: ☐

Email: eeeuser@gmail.com
Language: -- None --
Calendar integration: Outlook
Time zone: System (America/Los Angeles)
Date format: System (2026-08-19)
Business phone:
Mobile phone:
Photo: Click to add...

Update Set Password Delete

Related Links
[View linked accounts](#)
[View Subscriptions](#)
[Reset a password](#)

Role Creation

1. Navigate to User Administration → Roles.
2. Create the following roles:
 - a. bb1
 - b. bb2
 - c. bb3
 - d. bb4
3. Assign all four roles to EEE User.

Screenshot: Role Creation and Role Assignment.

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3Db2be81bcc320f10949353f1b4013125%26sysparm_record_target%3Dsys_user...

servicenow

AllFavoritesHistoryWorkspaces

Role - bb1

Search

Role: bb1

UpdateDelete

Name: bb1

Application: Global

Elevated privilege: ☐

Description:

UpdateDelete

Contains RolesApplications with RoleModules with RoleCustom Tables

for text Search

Role = bb1

Contains

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_user_role_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOToName%253e%253dbb%26syspa...

servicenow

AllFavoritesHistoryWorkspaces

Roles

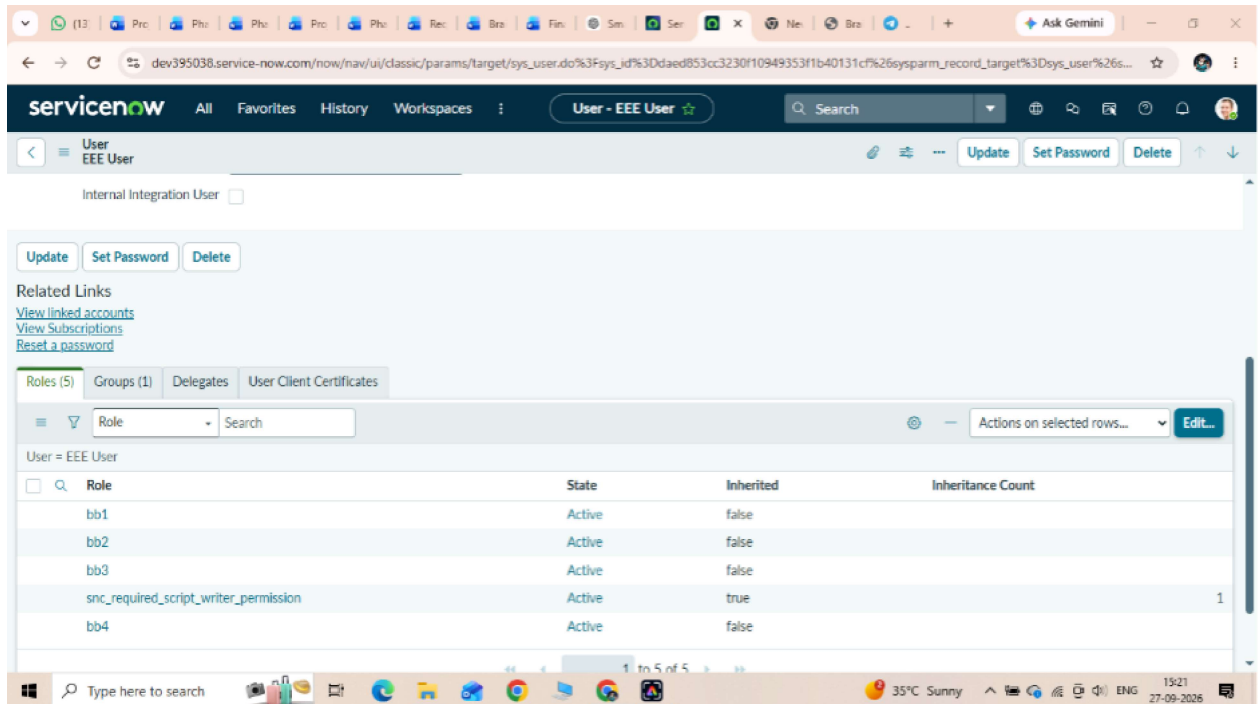
Search

Actions on selected rows... New

All > Name == bb

Name	Description	Elevated privilege
Search	Search	Search
bb1		false
bb2		false
bb3		false
bb4		false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note		

1 to 20 of 700



Milestone 2 – Institution Details Table Creation

1. Navigate to System Definition → Tables.
2. Create the custom table Institution Details (u_institution_details).
3. Add the following fields:
 - a. Student Roll Number (Auto Number)
 - b. Student Name (Reference)
 - c. Faculty Name (Reference)
 - d. Branch (Choice: ECE, EEE, CSE)
 - e. Email (String)
 - f. Phone Number (String)
 - g. Description (Multi-line String)
4. Save the table.
5. Create sample records for ECE, EEE, and CSE branches.

Screenshot: Institution Details Table and Sample Records.

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_db_object_list.do%3Fsysparm_userpref_module%3D7e7ca89ac0a8000901594ba32405461%26sysparm_q...

servicenow All Favorites History Workspaces Tables Search

Table Institution Details Delete Update Delete All Records

* Label Institution Details
* Name u_institution_details
Extends table

Application Global
Create module ☒
Create mobile module ☒
Add module to menu -- Create new --
New menu name
Remote Table ☐

Columns Controls Application Access

Table Columns for text Search New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false

Type here to search 35°C Sunny 14:43 27-09-2026

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D2ae8930c3630f10949353f1b4013106%26sysparm_record_target%3Dsys_db_...

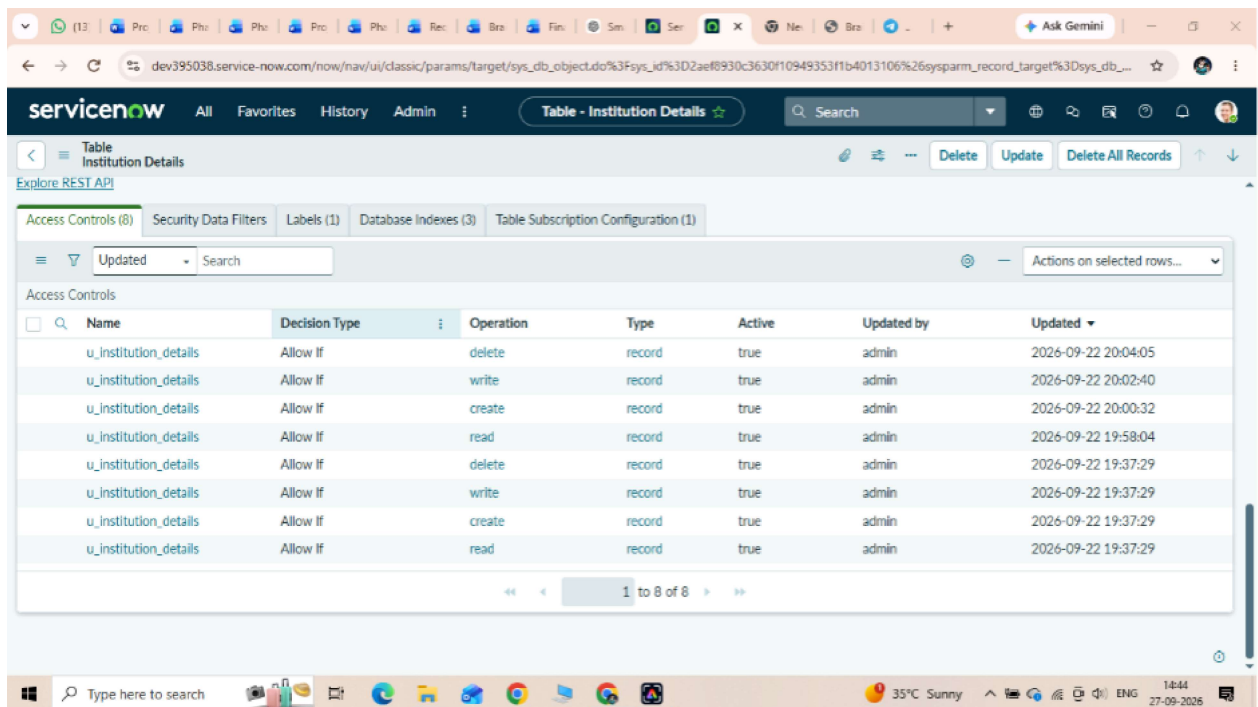
servicenow All Favorites History Admin Table - Institution Details Search

Table Institution Details Delete Update Delete All Records

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Student Roll Number	String	(empty)		40	false
Faculty Name	Reference	User		32	false
Updated	Date/Time	(empty)		40	false
Student Name	Reference	User		32	false
Branch	Choice	(empty)		40	false
Created by	String	(empty)		40	false
Email	String	(empty)		40	false
Description	String	(empty)		40	false
Phone Number	String	(empty)		40	false
Created	Date/Time	(empty)		40	false
Insert a new row...					

Type here to search 35°C Sunny 14:58 27-09-2026



Milestone 3 – Read ACL Demonstration

1. Navigate to System Security → Access Control (ACL).
2. Elevate the Security_admin role.
3. Create a Read ACL with:
 - a. Type: Record
 - b. Operation: Read
 - c. Name: u_institution_details
 - d. Requires Role: bb1
 - e. Data Condition: Branch = EEE
4. Add the JavaScript script.
5. Save the ACL.
6. Impersonate EEE User and open u_institution_details.list.
7. Verify that only EEE records are visible.
8. Impersonate Admin and verify that all records are visible.

Screenshots:

- Read ACL

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D24e31d38c3630f10949353f1b4013104%26sysparm_record_target%3Ds...

servicenow All Favorites History Access Control - u_institution_details Search

Access Control
u_institution_details

Type record Application Global

Operation read Active ☒

Admin overrides ☒ Advanced ☒

Decision Type Allow If

Protection policy -- None --

Name u_institution_details

Description Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE*EQ) evaluates to true.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role 1 to 1 of 1

- EEE User View

dev395038.service-now.com/now/nav/ui/classic/params/target/u_institution_details.list.do%3Fsysparm_clear_stack%3Dtrue

servicenow All Favorites History Institution Details Search Actions on selected rows... New

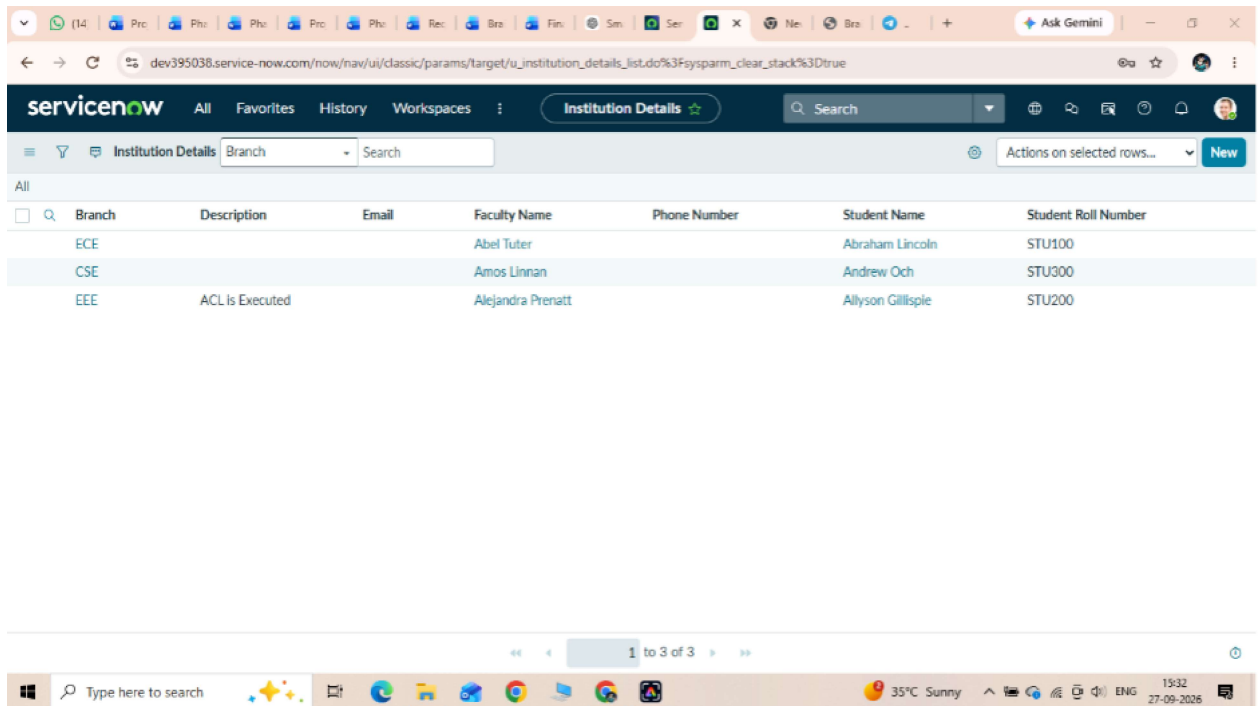
All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL Is Executed		Alejandra Prenatt		Allyson Gillisple	STU200

1 to 1 of 1

32°C Sunny 10:28 27-09-2026

- Admin View



Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
ECE			Abel Tuter		Abraham Lincoln	STU100
CSE			Amos Linnan		Andrew Och	STU300
EEE	ACL is Executed		Alejandra Prenatt		Allyson Gillisple	STU200

Milestone 4 – Create ACL Demonstration

1. Create a Create ACL.
2. Configure:
 - a. Type: Record
 - b. Operation: Create
 - c. Name: u_institution_details
 - d. Requires Role: bb2
3. Save the ACL.
4. Impersonate EEE User.
5. Verify that the New button appears.

Screenshot: New Button.

Milestone 5 – Write ACL Demonstration

1. Create a Write ACL.
2. Configure:
 - a. Type: Record
 - b. Operation: Write
 - c. Name: u_institution_details
 - d. Requires Role: bb3
3. Save the ACL.
4. Open an EEE record.
5. Edit the record and save it.

Screenshot: Edit Record.

Milestone 6 – Delete ACL Demonstration

1. Create a Delete ACL.
2. Configure:
 - a. Type: Record
 - b. Operation: Delete
 - c. Name: u_institution_details
 - d. Requires Role: bb4
3. Save the ACL.
4. Open an EEE record.
5. Verify that the Delete option is available.

Screenshot: Delete Option.

Final Verification

User	Permission
Admin	View, Create, Edit, Delete all records
bb1	View only EEE records
bb2	Create new records
bb3	Edit EEE records
bb4	Delete EEE records

Outcome

The demonstration successfully verified all six milestones. The ServiceNow application correctly enforced Read, Create, Write, and Delete permissions using Script-Controlled ACLs, ensuring that EEE branch records were accessible only to authorized users while administrators retained complete control over the system.

Conclusion

The Script-Controlled ACL – Restrict Record Access Based on Field Value project was successfully completed using the ServiceNow platform. The project demonstrated how Access Control Lists (ACLs) can be used to secure records by combining user roles with the Branch field value. A custom Institution Details

table was created, and users, roles, and Read, Create, Write, and Delete ACLs were configured to enforce role-based access.

The testing results confirmed that users with the appropriate roles could perform only their permitted operations, while administrators retained full access to all records. This project improved our understanding of role-based security, record-level access control, ACL scripting, and ServiceNow application development, providing practical experience in implementing secure access management.